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TECHNOLOGIES FOR DYNAMIC BATCH SIZE MANAGEMENT

Abstract

Technologies for dynamically managing a batch size of packets include a network device. The network device is to receive, into a queue, packets from a remote node to be processed by the network device, determine a throughput provided by the network device while the packets are processed, determine whether the determined throughput satisfies a predefined condition, and adjust a batch size of packets in response to a determination that the determined throughput satisfies a predefined condition. The batch size is indicative of a threshold number of queued packets required to be present in the queue before the queued packets in the queue can be processed by the network device.

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Background/Summary

CLAIM OF PRIORITY [0001] This application is a continuation of prior co-pending U.S. patent application Ser. No. 18/228,420, filed Jul. 31, 2023 and titled “TECHNOLOGIES FOR DYNAMIC BATCH SIZE MANAGEMENT,” which is a continuation of prior U.S. patent application Ser. No. 17/838,872, filed Jun. 13, 2022 and titled “TECHNOLOGIES FOR DYNAMIC BATCH SIZE MANAGEMENT,” now U.S. Pat. No. 11,757,802 issued on Sep. 12, 2023, which is a continuation of prior U.S. patent application Ser. No. 15/640,258, filed Jun. 30, 2017 and titled “TECHNOLOGIES FOR DYNAMIC BATCH SIZE MANAGEMENT,” now U.S. Pat. No. 11,362,968 issued on Jun. 14, 2022. Each of the aforesaid prior patent applications is hereby incorporated herein by reference in its entirety.

BACKGROUND
[0002] In recent years, ensuring high performance for software packet processing on computer architecture became paramount due to the fast development of network functions virtualization (NFV)/software-defined networking (SDN) and many new usage models such as telecommunication and Internet of Things (IoT) usages. Many efforts have been made to optimize throughput performance. Another important aspect of packet processing is latency or jitter, which may affect response time.
[0003] To optimize throughput performance, packets in a queue are processed in a batched fashion. Typically, an increase in a batch size of packets increases throughput performance. However, the increase in the batch size may negatively affect the latency and jitter performance by forcing packets to wait in the queue while gathering enough packets to form a batch of the predefined batch size. The negative impact may amplify when there are multiple processing stages. On the other hand, decreasing the batch size may decrease the latency performance but may negatively affect the throughput performance.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0004] The concepts described herein are illustrated by way of example and not by way of limitation in the accompanying figures. For simplicity and clarity of illustration, elements illustrated in the figures are not necessarily drawn to scale. Where considered appropriate, reference labels have been repeated among the figures to indicate corresponding or analogous elements.
[0005] FIG. **1** is a simplified block diagram of at least one embodiment of a system that includes a network device for dynamic batch size management;
[0006] FIG. **2** is a simplified block diagram of at least one embodiment of a network device of the system of FIG. **1**;
[0007] FIG. **3** is a simplified block diagram of at least one embodiment of an environment that may

be established by the network device of FIGS. **1** and **2**;

[0008] FIGS. **4-5** are a simplified flow diagram of at least one embodiment of a method for dynamically adjusting a batch size of packets to achieve an optimal combination of packet latency and throughput that may be executed by the network device of FIGS. **1-3**; and

[0009] FIG. **6** is a simplified diagram of at least one embodiment of the network device of FIGS. **1-3** in which the network device coordinates batching configurations across multiple processing stages.

DETAILED DESCRIPTION OF THE DRAWINGS

[0010] While the concepts of the present disclosure are susceptible to various modifications and alternative forms, specific embodiments thereof have been shown by way of example in the drawings and will be described herein in detail. It should be understood, however, that there is no intent to limit the concepts of the present disclosure to the particular forms disclosed, but on the contrary, the intention is to cover all modifications, equivalents, and alternatives consistent with the present disclosure and the appended claims.

[0011] References in the specification to “one embodiment,” “an embodiment,” “an illustrative embodiment,” etc., indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may or may not necessarily include that particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it is submitted that it is within the knowledge of one skilled in the art to effect such feature, structure, or characteristic in connection with other embodiments whether or not explicitly described. Additionally, it should be appreciated that items included in a list in the form of “at least one of A, B, and C” can mean (A); (B); (C); (A and B); (A and C); (B and C); or (A, B, and C). Similarly, items listed in the form of “at least one of A, B, or C” can mean (A); (B); (C); (A and B); (A and C); (B and C); or (A, B, and C).

[0012] The disclosed embodiments may be implemented, in some cases, in hardware, firmware, software, or any combination thereof. The disclosed embodiments may also be implemented as instructions carried by or stored on one or more transitory or non-transitory machine-readable (e.g., computer-readable) storage media, which may be read and executed by one or more processors. A machine-readable storage medium may be embodied as any storage device, mechanism, or other physical structure for storing or transmitting information in a form readable by a machine (e.g., a volatile or non-volatile memory, a media disc, or other media device).

[0013] In the drawings, some structural or method features may be shown in specific arrangements and/or orderings. However, it should be appreciated that such specific arrangements and/or orderings may not be required. Rather, in some embodiments, such features may be arranged in a different manner and/or order than shown in the illustrative figures. Additionally, the inclusion of a structural or method feature in a particular figure is not meant to imply that such feature is required in all embodiments and, in some embodiments, may not be included or may be combined with other features.

[0014] Referring now to FIG. **1**, in an illustrative embodiment, a system **100** for dynamically adjusting a batch size of packets to achieve an optimal combination of packet latency and throughput includes a source endpoint node **102** and a destination endpoint node **108** in communication over a network **104** via one or more network devices **106**. In use, as described in more detail below, the network device **106** dynamically adjusts the batch size (e.g., a threshold number of packets to be present in a queue before being processed by the network device **106**) to minimize packet latency while achieving a maximum throughput performance between the source endpoint node **102** and the destination endpoint node **108** over the network **104**. To do so, the network device **106** continuously or periodically monitors and adjusts the batch size of packets to achieve an optimal batching configuration by balancing the throughput performance and the latency performance. The large batch size of packets may maximize the throughput performance but may negatively affect the latency and jitter performance, while a relatively small batch size of packets may improve the latency performance (e.g., reduce latency) but may negatively affect the throughput performance (e.g., reduce throughput).

[0015] In operation, a network device (e.g., the network device **106**) may process packets in a single processing stage or in multiple processing stages. Each processing stage, in the illustrative embodiment, has its own queue, batch size of packets, and throughput. In the illustrative embodiment, the network device **106** dynamically adjusts the batch size for each processing stage based on a throughput performance of the corresponding processing stage and a queue condition for the corresponding processing stage (e.g., a number of packets in the queue and an increasing or decreasing rate of change in packets in the queue). For example, the network device **106** may increase the batch size of packets to increase the throughput performance for the processing stage until the throughput performance no longer increases, indicating that further increasing the batch size will only negatively affect the latency without increasing the throughput. Additionally, the network device **106** may further synchronize the batch size across the multiple processing stages in the same processing pipeline as discussed in detail in FIG. **5**.

[0016] The source endpoint node **102** may be embodied as any type of computation or computing device capable of performing the functions described herein, including, without limitation, a computer, a desktop computer, a smartphone, a workstation, a laptop computer, a notebook computer, a tablet computer, a mobile computing device, a wearable computing device, a network appliance, a web appliance, a distributed computing system, a processor-based system, and/or a consumer electronic device. Similarly, the destination endpoint node **108** may be embodied as any type of computation or computing device capable of performing the functions described herein, including, without limitation, a computer, a desktop computer, a smartphone, a workstation, a laptop computer, a notebook computer, a tablet computer, a mobile computing device, a wearable computing device, a network appliance, a web appliance, a distributed computing system, a processor-based system, and/or a consumer electronic device. Each of the source endpoint node **102** and the destination endpoint node **108** may include components commonly found in a computing device such as a processor, memory, input/output subsystem, data storage, communication circuitry, etc.

[0017] The network **104** may be embodied as any type of wired or wireless communication network, including cellular networks (e.g., Global System for Mobile Communications (GSM), 3G, Long Term Evolution (LTE), Worldwide Interoperability for Microwave Access (WiMAX), etc.), digital subscriber line (DSL) networks, cable networks (e.g., coaxial networks, fiber networks, etc.), telephony networks, local area networks (LANs) or wide area networks (WANs), global networks (e.g., the Internet), or any combination thereof. Additionally, the network **104** may include any number of network devices **106** as needed to facilitate communication between the source endpoint node **102** and the destination endpoint node **108**.

[0018] Each network device **106** may be embodied as any type of computing device capable of facilitating wired and/or wireless network communications between the source endpoint node **102** and the destination endpoint node **108**. For example, each network device **106** may be embodied as a server (e.g., stand-alone, rack-mounted, blade, etc.), a router, a switch, a network hub, an access point, a storage device, a compute device, a multiprocessor system, a network appliance (e.g., physical or virtual), a computer, a desktop computer, a smartphone, a workstation, a laptop computer, a notebook computer, a tablet computer, a mobile computing device, or any other computing device capable of processing network packets.

[0019] As shown in FIG. **2**, an illustrative network device **106** includes a central processing unit (CPU) **210**, a main memory **212**, an input/output (I/O) subsystem **214**, communication circuitry **216**, and a data storage device **220**. Of course, in other embodiments, the network device **106** may include other or additional components, such as those commonly found in a computer (e.g., data storage, display, etc.). Additionally, in some embodiments, one or more of the illustrative components may be incorporated in, or otherwise form a portion of, another component. For example, in some embodiments, the main memory **212**, or portions thereof, may be incorporated in the CPU **210**.

[0020] The CPU **210** may be embodied as any type of processor capable of performing the functions described herein. The CPU **210** may be embodied as a single or multi-core processor(s), a microcontroller, or other processor or processing/controlling circuit. In some embodiments, the CPU **210** may be embodied as, include, or be coupled to a field programmable gate array (FPGA), an application specific integrated circuit (ASIC), a graphics processing unit (GPU), reconfigurable hardware or hardware circuitry, or other specialized hardware to facilitate performance of the functions described herein. In the illustrative embodiment, the CPU **210** is embodied as a processor containing a set **230** of multiple cores **232**, **234**, **236**, **238**, **240**, **242**, **244**, and **246**. While eight cores are shown in FIG. **2**, it should be understood that in other embodiments, the CPU **210** may contain a different number of cores. Similarly, the main memory **212** may be embodied as any type of volatile (e.g., dynamic random access memory (DRAM), etc.) or non-volatile memory or data storage capable of performing the functions described herein. In some embodiments, all or a portion of the main memory **212** may be integrated into the CPU **210**. In operation, the main memory **212** may store various data and software used during operation of the network device **106** such as packet data, operating systems, applications, programs, libraries, and drivers.

[0021] In addition, in some embodiments, the CPU **210** may also include a batch size manager logic unit **250**. The batch size manager logic unit **250** may be embodied as any hardware device (e.g., a co-processor, an FPGA, and ASIC, or circuitry) capable of performing functions that include dynamic batch size adjustment in a queue environment for each processing stage. More specifically, the batch size manager logic unit **250** is any device capable of performing the batch size adjustment described with respect to FIGS. **4-6** below. In some embodiments, the batch size manager logic unit **250** may be virtualized.

[0022] The I/O subsystem **214** may be embodied as circuitry and/or components to facilitate input/output operations with the CPU **210**, the main memory **212**, and other components of the network device **106**. For example, the I/O subsystem **214** may be embodied as, or otherwise include, memory controller hubs, input/output control hubs, integrated sensor hubs, firmware devices, communication links (i.e., point-to-point links, bus links, wires, cables, light guides, printed circuit board traces, etc.), and/or other components and subsystems to facilitate the input/output operations. In some embodiments, the I/O subsystem **214** may form a portion of a system-on-a-chip (SoC) and be incorporated, along with one or more of the CPU **210**, the main memory **212**, and other components of the network device **106**, on a single integrated circuit chip.

[0023] The communication circuitry **216** may be embodied as any communication circuit, device, or collection thereof, capable of enabling communications over the network **104** between the network device **106** and the source endpoint node **102**, another network device **106**, and/or the destination endpoint node **108**. The communication circuitry **216** may be configured to use any one or more communication technology (e.g., wired or wireless communications) and associated protocols (e.g., Ethernet, Bluetooth®, Wi-Fi®, WiMAX, etc.) to effect such communication.

[0024] The illustrative communication circuitry **216** includes a network interface controller (NIC) **218**, which may also be referred to as a host fabric interface (HFI). The NIC **218** may be embodied as one or more add-in-boards, daughtercards, network interface cards, controller chips, chipsets, or other devices that may be used by the network device **106** to connect the source endpoint node **102**, the destination endpoint node **108**, and/or another network device **106**. In some embodiments, the NIC **218** may be embodied as part of a system-on-a-chip (SoC) that includes one or more processors, or included on a multichip package that also contains one or more processors. In some embodiments, the NIC **218** may include a local processor (not shown) and/or a local memory (not shown) that are both local to the NIC **218**. In such embodiments, the local processor of the NIC **218** may be capable of performing one or more of the functions of the CPU **210** described herein.

Additionally or alternatively, in such embodiments, the local memory of the NIC **218** may be integrated into one or more components of the network device **106** at the board level, socket level, chip level, and/or other levels.
[0025] The data storage device **220** may be embodied as any type of device or devices configured for short-term or long-term storage of data such as, for example, memory devices and circuits, memory cards, hard disk drives, solid-state drives, or other data storage devices. The data storage device **220** may include a system partition that stores data and firmware code for the network device **106**. The data storage device **220** may also include an operating system partition that stores data files and executables for an operating system of the network device **106**.
[0026] Additionally, the network device **106** may include one or more peripheral devices **224**. Such peripheral devices **224** may include any type of peripheral device commonly found in a compute device such as speakers, a mouse, a keyboard, and/or other input/output devices, interface devices, and/or other peripheral devices. It should be appreciated that the network device **106** may include other components, sub-components and/or devices commonly found in a network device, which are not illustrated in FIG. **2** for clarity of the description.
[0027] It should be appreciated that, in some embodiments, the network device **106** may be embodied as a core of a central processing unit (CPU) of a compute device that is capable of communicating between the source end point **102** and the destination end point **108**. In such embodiments, the CPU of the compute device may include a batch size manager logic unit similar to the batch size manager logic unit **250**, capable of performing the functions described above.
[0028] Referring now to FIG. **3**, in the illustrative embodiment, each network device **106** may establish an environment **300** during operation. The illustrative environment **300** includes a network communication manager **320**, a packet processor **330**, a queue condition monitor **340**, a throughput monitor **350**, and a batch size adjuster **360**, which includes a batch size increaser **362**, a batch size decreaser **364**, and a batch size synchronizer **366**. Each of the components of the environment **300** may be embodied as hardware, firmware, software, or a combination thereof. As such, in some embodiments, one or more of the components of the environment **300** may be embodied as circuitry or collection of electrical devices (e.g., a network communication manager circuit **320**, a packet processor circuit **330**, a queue condition monitor circuit **340**, a throughput monitor circuit **350**, a batch size adjuster circuit **360**, a batch size increaser circuit **362**, a batch size decreaser circuit **364**, a batch size synchronizer circuit **366**, etc.). It should be appreciated that, in such embodiments, one or more of the network communication manager circuit **320**, the packet processor circuit **330**, the queue condition monitor circuit **340**, the throughput monitor circuit **350**, the batch size adjuster circuit **360**, the batch size increaser circuit **362**, the batch size decreaser circuit **364**, or the batch size synchronizer circuit **366** may form a portion of one or more of the CPU **210**, the main memory **212**, the I/O subsystem **214**, the communication circuitry **216** and/or other components of the network device **106**. Additionally, in some embodiments, one or more of the illustrative components may form a portion of another component and/or one or more of the illustrative components may be independent of one another. Further, in some embodiments, one or more of the components of the environment **300** may be embodied as virtualized hardware components or emulated architecture, which may be established and maintained by the CPU **210** or other components of the network device **106**.
[0029] The network communication manager **320**, which may be embodied as hardware, firmware, software, virtualized hardware, emulated architecture, and/or a combination thereof as discussed above, is configured to facilitate inbound and outbound network communications (e.g., network traffic, network packets, network flows, etc.) to and from the network device **106**, respectively. To do so, the network communication manager **320** is configured to receive and process data packets from one computing device (e.g., the source endpoint node **102**, another network device **106**, the destination endpoint node **108**) and to prepare and send data packets to another computing device (e.g., the source endpoint node **102**, another network device **106**, the destination endpoint node

108). Accordingly, in some embodiments, at least a portion of the functionality of the network communication manager **320** may be performed by the communication circuitry **216**, and, in the illustrative embodiment, by the NIC **218**.

[0030] The packet processor **330**, which may be embodied as hardware, firmware, software, virtualized hardware, emulated architecture, and/or a combination thereof as discussed above, is configured to process (e.g., perform operations on data in the packets, such as compression, decompression, encryption, decryption, and/or other operations) one or more packets in the queue. As described above, in the illustrative embodiment, the packets in the queue are processed in a batched fashion to maximize the throughput. The packet processor **330** may commence to process the packets when a threshold number of packets (i.e., a batch size of the processing stage) are present in the queue of the processing stage.

[0031] The queue condition monitor **340**, which may be embodied as hardware, firmware, software, virtualized hardware, emulated architecture, and/or a combination thereof as discussed above, is configured to monitor the queue condition(s) of the processing stage(s). For example, the queue condition may include a number of packets present in the queue and an increasing or decreasing rate of change in the number of packets in the queue.

[0032] The throughput monitor **350**, which may be embodied as hardware, firmware, software, virtualized hardware, emulated architecture, and/or a combination thereof as discussed above, is configured to monitor the throughput of each processing stage. As described above, the overall network throughput is a rate at which the packet data is being transmitted or delivered from the source endpoint node **102** to the destination endpoint node **108** over the network **104**. When the packets are processed in multiple processing stages, the throughput of each processing stage is a rate at which the packet data is being processed and forwarded to a next processing stage. As such, the throughput of each processing stage may be monitored and adjusted to optimize the overall network throughput.

[0033] The batch size adjuster **360**, which may be embodied as hardware, firmware, software, virtualized hardware, emulated architecture, and/or a combination thereof as discussed above, is configured to adjust a batch size for each processing stage based on the queue condition and the throughput of each processing stage. To do so, in the illustrative embodiment, the batch size adjuster **360** includes the batch size increaser **362**, batch size decreaser **364**, and the batch size synchronizer **366**.

[0034] The batch size increaser **362**, which may be embodied as hardware, firmware, software, virtualized hardware, emulated architecture, and/or a combination thereof as discussed above, is configured to increase the batch size for the processing stage to increase the throughput of the processing stage. In the illustrative embodiment, the batch size increaser **362** may multiplicatively increase the batch size for the processing stage using an additive decrease and multiplicative increase algorithm to quickly increase the batch size to avoid any packet loss due to packet congestion and to probe for an optimal batch size to decrease the latency performance while maintaining the throughput. If the batch size becomes too large, it negatively affects the latency performance by forcing the packets to wait in the queue while gathering enough packets to reach the batch size.

[0035] The batch size decreaser **364**, which may be embodied as hardware, firmware, software, virtualized hardware, emulated architecture, and/or a combination thereof as discussed above, is configured to decrease the batch size for the processing stage to decrease the latency of the processing stage. In the illustrative embodiment, the batch size decreaser **364** may additively decrease the batch size for the processing stage using the additive decrease and multiplicative increase algorithm to decrease the batch size slowly to probe for an optimal batch size to decrease the latency performance while maintaining the throughput. In other words, the batch size is gradually decreased to reach a threshold where the throughput is maintained based on the queuing condition of the processing stage (e.g., the rate at which incoming packets are received). If the batch size is decreased below the threshold, the throughput will be negatively affected (e.g., decreased) and incoming packets may be dropped.

[0036] The batch size synchronizer **366**, which may be embodied as hardware, firmware, software, virtualized hardware, emulated architecture, and/or a combination thereof as discussed above, is configured to synchronize the batch size across the multiple processing stages in the same processing pipeline. To do so, the batch size synchronizer **366** compares the present batch size for the present processing stage to batch size(s) for the other processing stage(s) to determine which batch size is greater. The batch size synchronizer **366** may synchronize the present batch size for the present processing stage if the present batch size is smaller than the batch size(s) for the other processing stage(s) to avoid any packet loss due to congestion. In such case, even if the synchronized batch size is larger than the optimal batch size for the processing stage, the larger synchronized batch size ensures that the packets are not dropped while being forwarded to the next processing stage. However, if the present batch size for the present processing stage is greater than the batch size(s) for the other processing stage(s), the batch size synchronizer **366** does not synchronize the present batch size for the present processing stage.

[0037] In the illustrative environment **300**, the network device **106** also includes packet data **302**, throughput data **304**, and batch size data **306**. The packet data **302** may be embodied as any data that is indicative of packet payloads, packet source and destination information, and packet metadata. The metadata defines properties of a packet, such as the packet size, an input port number, an output port number, and state data that is indicative of which processing stages have completed processing of the data packet and whether the data packet is ready for transmission to another device (e.g., to the destination endpoint node **108**).

[0038] The throughput data **304** may be embodied as any data that is indicative of throughput for each processing stage. As discussed above, the throughput data **304** may be used to optimize the batch size of each processing stage across the multiple processing stages.

[0039] The batch size data **306** may be embodied as any data that is indicative of a batch size for a corresponding processing stage. As discussed above, each processing stage has its own batch size of packets, and the batch size for each processing stage may be different. In such case, the batching may be synchronized across the processing stage(s) to minimize the latency while achieving the maximum overall throughput of the network device **106**.

[0040] Referring now to FIG. **4**, in use, the network device **106** may execute a method **400** for dynamic batch size management for each processing stage. In the illustrative embodiment, the network device **106** continuously or periodically monitors the throughput and queuing condition for the present processing stage and determines an optimal batching configuration to minimize the latency performance while maintaining maximum throughput. It should be appreciated that the batching configuration may be adjusted based on throughputs and queuing conditions of multiple processing stages in the same processing pipeline to achieve the overall best performance. When one stage has egress interfaces with multiple queues for multiple processing stages, the network device **106** monitors each queue condition independently and makes the decision on the optimal configuration for each stage accordingly. Subsequently, the batch sizes of the processing stages are configured to be synchronized across the multiple processing stages.

[0041] The method **400** begins with block **402**, in which the network device **106** determines whether to perform the batch size management. If the network device **106** determines not to perform the batch size management, the method **400** loops back to block **402** to continue determining whether to perform the batch size management. If, however, the network device **106** determines to perform the batch size management, the method **400** advances to block **404**. In some embodiments, the network device **106** may determine to perform the batch size management in response to determining that the batch size manager logic unit **250** is present, in response to a determination that a setting stored in a configuration file in the data storage device **220** indicates to perform the batch size management, and/or as a function of other criteria.

[0042] In block **404**, the network device **106** processes packets present in a queue of a present processing stage. As discussed above, in the illustrative embodiment, the network device **106** processes the packets over multiple processing stages. In some embodiments, however, only one processing stage may be used to process the incoming packets.
[0043] To do so, in some embodiments in block **406**, the network device **106** may receive a packet within the present processing stage. In block **408**, the network device **106** may perform a deep packet inspection of the received packet to inspect the packet data for candidate patterns and take actions based on the presence or absence of these patterns (e.g., searching for protocol non-compliance, viruses, spam, intrusions, or defined criteria to decide whether the packet may pass or if the packet should be routed to a different destination). In some embodiments, the network device **106** may compress or decompress the packet data and/or encrypt or decrypt the packet data, as indicated in blocks **410** and **412**. It should be appreciated that the network device **106** may further perform other processing of the packet data in block **414**.
[0044] In block **416**, the network device **106** determines a present throughput for the present processing stage. As discussed above, the network throughput is a rate at which the packet data is being transmitted or delivered from the source endpoint node **102** to the destination endpoint node **108** over the network **104**. When the packets are processed in multiple processing stages of the pipeline, the network throughput may be determined based on throughputs for each processing stages. In other words, the present throughput for the present processing stage is indicative of a rate at which the network device **106** is processing the packet data in the present processing stage.
[0045] In block **418**, the network device **106** determines whether the determined present throughput for the present processing stage is insufficient. In other words, the network device **106** determines whether the present throughput satisfies a predefined condition. For example, as described above, the network device **106** may increase the batch size of packets to increase the throughput performance for the processing stage until the throughput performance no longer increases (i.e., the predefined condition), indicating that further increasing the batch size will only negatively affect the latency without increasing the throughput. As such, the network device **106** may determine whether the present throughput for the present processing stage has satisfied the predefined condition.
[0046] To do so, in some embodiments in block **420**, the network device **106** determines whether a number of packets in a queue for the present processing stage is increasing to infer the throughput performance. For example, if the number of packets in the queue for the present processing stage continues to increase, it means that there are more packets coming in to the queue than packets being processed. As such, the increase in the number of packets in the queue indicates that the present throughput should be increased to maximize the throughput. On the other hand, if the number of packets in the queue starts to decrease, it means that there are more packets being processed than packets coming in to the queue. As such, the decrease in the number of packets in the queue indicates that the present batch size could be decreased to decrease the latency while maintaining the present throughput.
[0047] If the network device **106** determines that the present throughput for the present processing stage is not insufficient in block **422**, the method **400** advances to block **424**. For example, the network device **106** may determine that further increasing the batch size will only negatively affect the latency without increasing the throughput because the rate at which incoming packets are received in the queue has been reduced. In block **424**, the network device **106** decreases the batch size for the present processing stage. To do so, in some embodiments in block **426**, the network device **106** may additively decrease the batch size for the present processing stage using an additive decrease and multiplicative increase algorithm to decrease the batch size slowly to probe for an optimal batch size to decrease the latency performance while maintaining the throughput. In other words, the batch size is gradually decreased to reach a threshold where the throughput is maintained based on the queuing condition of the present processing stage (e.g., the rate at which

incoming packets are received). If the batch size is decreased below the threshold, it will start to negatively affect the throughput. In addition, a batch size that is too small may cause incoming packets to be dropped. In response to decreasing the batch size of packets for the present processing stage, the method **400** advances to block **432** in FIG. **5** to synchronize the batch size across the multiple processing stages in the same processing pipeline.

[0048] Referring back to block **422**, if the network device **106** determines that the present throughput for the present processing stage is insufficient, the method **400** advances to block **428**. In block **428**, the network device **106** increases the batch size for the present processing stage. To do so, in some embodiments in block **430**, the network device **106** may multiplicatively increase the batch size for the present processing stage using an additive decrease and multiplicative increase algorithm to quickly increase the batch size to avoid any packet loss due to packet congestion and to probe for an optimal batch size to improve the latency performance (e.g., decrease the latency) while maintaining the throughput. However, if the batch size becomes too large, it negatively affects the latency performance by forcing the packets to wait in the queue while gathering enough packets to reach the batch size. In response to increasing the batch size of packets for the present processing stage, the method **400** advances to block **432** in FIG. **5** to synchronize the batch size across the multiple processing stages in the same processing pipeline.

[0049] In block **432**, the network device **106** compares the batch size for the present processing stage to every batch size for preceding processing stage(s). In block **434**, the network device **106** determines whether the batch size for the present processing stage is greater than the batch size for any one of the preceding processing stages. If the network device **106** determines that the batch size of the present processing stage is not greater than the batch size for any one of the preceding processing stages, network device **106** maintains the present batch size and the method **400** skips ahead to block **440** in which the network device **106** sets the present processing stage to the next stage (e.g., the network device **106** proceeds to the next processing stage for analysis).

[0050] The method **400** then loops back to block **404** to continuously or periodically monitor and adjust the optimal batch size for the present processing stage to minimize the latency while maintaining the throughput to achieve the overall best performance of the network device **106**.

[0051] If, however, the network device **106** determines that the batch size of the present processing stage is greater than all the batch size(s) for the preceding processing stage(s), the method **400** advances to block **436**. In block **436**, the network device **106** synchronizes the batch size across the processing stage(s). To do so, in some embodiments in block **438**, the network device **106** may set the batch size of the preceding processing stage(s) to the batch size for the present processing stage. This may allow the network device **106** to update an overall optimal batch size across the multiple processing stages to achieve the overall best performance of the network device **106**.

[0052] The method **400** then advances to block **440** in which the network device **106** sets the present processing stage to the next stage and loops back to block **404** to continuously or periodically monitor and adjust the optimal batch size for the present processing stage to minimize the latency while maintaining the throughput. While the method **400** is described above as operating on each processing stage sequentially, in other embodiments, the network device **106** may operate on each processing stage concurrently (e.g., in separate threads) and share data (e.g., batch size information) across the threads.

[0053] Referring now to FIG. **6**, in an illustrative embodiment, the network device **106** is configured to coordinate batching configurations across the multiple processing stages in the same processing pipeline. As shown in FIG. **6**, there are n processing stages that process incoming packets. The incoming packets are received in a stage 1 **602** from a source endpoint node **102**. The packets wait in a queue of the stage 1 **602** while gathering enough packets to form a batch of predefined batch size for the stage 1 **602**. As discussed in detail above, the network device **106** determines an optimal batching configuration for the stage 1 **602** to minimize the latency while maintaining throughput for the stage 1 **602**. When enough packets are gathered in the queue of the stage 1 **602** to satisfy the optimal batching configuration for the stage 1 **602**, the batch of packets (i.e., batching 1) is forwarded to a stage 2 **604**.

[0054] Similarly, the stage 2 **604** has its own queue, batch size, and throughput. The network device **106** determines an optimal batching configuration for the stage 2 **604** to minimize the latency while maintaining the same throughput for the stage 2 **604**. In some embodiments, when enough packets are gathered in the queue of the stage 2 **604** to satisfy the optimal batching configuration for the stage 2 **604**, the batch of packets (i.e., batching 2) is forwarded to a next processing stage.

[0055] However, in the illustrative embodiment, the network device **106** may further synchronize the batch size for the stage 1 **602** and the stage 2 **604**. It should be appreciated that the network device **106** may only synchronize the batch size for the stage 2 **604** if the batch size for the stage 2 **604** is greater than the batch size for the stage 1 **602**. If the batch size for the stage 2 **604** is greater than the batch size for the stage 1 **602**, the network device **106** updates the overall optimal batch size across the multiple processing stages to the batch size for the stage 2 **604**. As discussed above, even if the synchronized batch size is larger than the optimal batch size for the stage 1 **602**, the larger batch size ensures that the packets are not dropped while being forwarded to the next processing stage. If, however, the batch size for the stage 2 **604** is smaller than the batch size for the stage 1 **602**, the network device **106** does not synchronize the batch size. When enough packets are gathered in the queue of the stage 2 **604** to satisfy the synchronized batch size for the stage 2 **604**, the batch of packets (i.e., batching 2) is forwarded to a next processing stage. The forwarding step is repeated until the packets reach a stage N **606** to be processed for the last time before transmitted to a destination endpoint node **108**.

[0056] In some embodiments, each stage may distribute its batch of packets to multiple egress queues, where each egress queue may be processed by a core. Each queue may correspond to a core for the next stage processing. For each core, a new pipeline batch size may be determined for each stage, and synchronization of batch sizes may be performed between the stages for each core in a similar manner as described above. For example, at stage 1, a new pipeline batch size may be determined for core 1 for stage 1. When the packets are forwarded to stage 2, core 1 may determine a new pipeline batch size for stage 2. The network device **106** may then synchronize the batch size for the stage 1 and the stage 2 for core 1 in the similar manner as described above such that the batch sizes of each core are synchronized.

Examples

[0057] Illustrative examples of the technologies disclosed herein are provided below. An embodiment of the technologies may include any one or more, and any combination of, the examples described below.

[0058] Example 1 includes a network device comprising a network interface controller; and one or more processors coupled to the network interface controller, wherein the one or more processors are to receive, into a queue, packets from a remote node to be processed by the network device, received via the network interface controller; determine a throughput provided by the network device while the packets are processed; determine whether the determined throughput satisfies a predefined condition; and adjust a batch size of packets in response to a determination that the determined throughput satisfies a predefined condition; wherein the batch size is indicative of a threshold number of queued packets required to be present in the queue before the queued packets in the queue can be processed by the network device.

[0059] Example 2 includes the subject matter of Example 1, and wherein to determine whether the determined throughput satisfies a predefined condition comprises to determine whether the number of packets in the queue has increased over time.

[0060] Example 3 includes the subject matter of any of Examples 1 and 2, and wherein to adjust the batch size comprises to increase the batch size.

[0061] Example 4 includes the subject matter of any of Examples 1-3, and wherein to increase the batch size comprises to multiplicatively increase the batch size for a processing stage of the network device.

[0062] Example 5 includes the subject matter of any of Examples 1-4, and wherein the one or more processors are further to adjust the batch size in response to a determination that the determined throughput does not satisfy the predefined condition.

[0063] Example 6 includes the subject matter of any of Examples 1-5, and wherein to adjust the batch size comprises to decrease the batch size for a processing stage of the network device.

[0064] Example 7 includes the subject matter of any of Examples 1-6, and wherein to adjust the batch size comprises to additively decrease the batch size for a processing stage of the network device.

[0065] Example 8 includes the subject matter of any of Examples 1-7, and wherein the one or more processors are further to process one or more packets with multiple processing stages, wherein each processing stage performs a different operation on packet data.

[0066] Example 9 includes the subject matter of any of Examples 1-8, and wherein the one or more processors are further to synchronize the adjusted batch size across multiple processing stages of the network device.

[0067] Example 10 includes the subject matter of any of Examples 1-9, and wherein to synchronize the adjusted batch size comprises to compare the batch size of a present processing stage with the batch size for every preceding processing stage in the network device; and adjust, in response to a determination that the batch size of the present processing stage is greater than the batch size for every preceding processing stage, the batch size for every preceding processing stage to the batch size of the present processing stage.

[0068] Example 11 includes the subject matter of any of Examples 1-10, and wherein the one or more processors are further to compare the batch size of a present processing stage with a batch size for a preceding processing stage; and maintain, in response to a determination that the batch size of the present processing stage is smaller than the batch size for the preceding processing stage, the batch size of the preceding processing stage.

[0069] Example 12 includes a method for dynamically managing a batch size of packets, the method comprising receiving, by a network device and into a queue, packets from a remote node to be processed by the network device, received via a network interface controller of the network device; determining, by the network device, a throughput provided by the network device while processing the packets; determining, by the network device, whether the determined throughput satisfies a predefined condition; and adjusting, by the network device, a batch size of packets in response to a determination that the determined throughput satisfies a predefined condition; wherein the batch size is indicative of a threshold number of queued packets required to be present in the queue before the queued packets in the queue can be processed by the network device.

[0070] Example 13 includes the subject matter of Example 12, and wherein determining whether the determined throughput satisfies a predefined condition comprises determining whether the number of packets in the queue has increased over time.

[0071] Example 14 includes the subject matter of any of Examples 12 and 13, and wherein adjusting the batch size comprises increasing the batch size.

[0072] Example 15 includes the subject matter of any of Examples 12-14, and wherein increasing the batch size comprises multiplicatively increasing the batch size for a processing stage of the network device.

[0073] Example 16 includes the subject matter of any of Examples 12-15, and further including adjusting, by the network device, the batch size in response to a determination that the determined throughput does not satisfy the predefined condition.

[0074] Example 17 includes the subject matter of any of Examples 12-16, and wherein adjusting the batch size comprises decreasing the batch size for a processing stage of the network device.

[0075] Example 18 includes the subject matter of any of Examples 12-17, and wherein adjusting the batch size comprises additively decreasing the batch size for a processing stage of the network device.

[0076] Example 19 includes the subject matter of any of Examples 12-18, and further including processing, by the network device, one or more packets with multiple processing stages, wherein each processing stage performs a different operation on packet data.

[0077] Example 20 includes the subject matter of any of Examples 12-19, and further including synchronizing, by the network device, the adjusted batch size across multiple processing stages of the network device.

[0078] Example 21 includes the subject matter of any of Examples 12-20, and wherein synchronizing the adjusted batch size comprises comparing, by the network device, the batch size of a present processing stage with the batch size for every preceding processing stage in the network device; and adjusting, by the network device and in response to a determination that the batch size of the present processing stage is greater than the batch size for every preceding processing stage, the batch size for every preceding processing stage to the batch size of the present processing stage.

[0079] Example 22 includes the subject matter of any of Examples 12-21, and further including comparing, by the network device, the batch size of a present processing stage with a batch size for a preceding processing stage; and maintaining, by the network device in response to a determination that the batch size of the present processing stage is smaller than the batch size for the preceding processing stage, the batch size of the preceding processing stage.

[0080] Example 23 includes one or more machine-readable storage media comprising a plurality of instructions stored thereon that, in response to being executed, cause a network device to perform the method of any of Examples 12-22.

[0081] Example 24 includes a network device comprising one or more processors; one or more memory devices having stored therein a plurality of instructions that, when executed by the one or more processors, cause the compute device to perform the method of any of Examples 12-22.

[0082] Example 25 includes a network device comprising means for performing the method of any of Examples 12-22.

[0083] Example 26 includes a network device comprising a network interface controller; and batch size adjuster circuitry to receive, into a queue, packets from a remote node to be processed by the network device, received via the network interface controller; determine a throughput provided by the network device while the packets are processed; determine whether the determined throughput satisfies a predefined condition; and adjust a batch size of packets in response to a determination that the determined throughput satisfies a predefined condition; wherein the batch size is indicative of a threshold number of queued packets required to be present in the queue before the queued packets in the queue can be processed by the network device.

[0084] Example 27 includes the subject matter of Example 26, and wherein to determine whether the determined throughput satisfies a predefined condition comprises to determine whether the number of packets in the queue has increased over time.

[0085] Example 28 includes the subject matter of any of Examples 26 and 27, and wherein to adjust the batch size comprises to increase the batch size.

[0086] Example 29 includes the subject matter of any of Examples 26-28, and wherein to increase the batch size comprises to multiplicatively increase the batch size for a processing stage of the network device.

[0087] Example 30 includes the subject matter of any of Examples 26-29, and wherein the batch size adjuster circuitry is further to adjust the batch size in response to a determination that the determined throughput does not satisfy the predefined condition.

[0088] Example 31 includes the subject matter of any of Examples 26-30, and wherein to adjust the batch size comprises to decrease the batch size for a processing stage of the network device.

[0089] Example 32 includes the subject matter of any of Examples 26-31, and wherein to adjust the

batch size comprises to additively decrease the batch size for a processing stage of the network device.
[0090] Example 33 includes the subject matter of any of Examples 26-32, and further including packet processor circuitry to process one or more packets with multiple processing stages, wherein each processing stage performs a different operation on packet data.
[0091] Example 34 includes the subject matter of any of Examples 26-33, and wherein the batch size adjuster circuitry is further to synchronize the adjusted batch size across multiple processing stages of the network device.
[0092] Example 35 includes the subject matter of any of Examples 26-34, and wherein to synchronize the adjusted batch size comprises to compare the batch size of a present processing stage with the batch size for every preceding processing stage in the network device; and adjust, in response to a determination that the batch size of the present processing stage is greater than the batch size for every preceding processing stage, the batch size for every preceding processing stage to the batch size of the present processing stage.
[0093] Example 36 includes the subject matter of any of Examples 26-35, and wherein the batch size adjuster circuitry is further to compare the batch size of a present processing stage with a batch size for a preceding processing stage; and maintain, in response to a determination that the batch size of the present processing stage is smaller than the batch size for the preceding processing stage, the batch size of the preceding processing stage.
[0094] Example 37 includes a network device comprising circuitry for receiving, into a queue, packets from a remote node to be processed by the network device, received via a network interface controller of the network device; circuitry for determining a throughput provided by the network device while processing the packets; circuitry for determining whether the determined throughput satisfies a predefined condition; and means for adjusting a batch size of packets in response to a determination that the determined throughput satisfies a predefined condition; wherein the batch size is indicative of a threshold number of queued packets required to be present in the queue before the queued packets in the queue can be processed by the network device.
[0095] Example 38 includes the subject matter of Example 37, and wherein circuitry for determining whether the determined throughput satisfies a predefined condition comprises circuitry for determining whether the number of packets in the queue has increased over time.
[0096] Example 39 includes the subject matter of any of Examples 37 and 38, and wherein the means for adjusting the batch size comprises means for increasing the batch size.
[0097] Example 40 includes the subject matter of any of Examples 37-39, and wherein the means for increasing the batch size comprises means for multiplicatively increasing the batch size for a processing stage of the network device.
[0098] Example 41 includes the subject matter of any of Examples 37-40, and further including means for adjusting the batch size in response to a determination that the determined throughput does not satisfy the predefined condition.
[0099] Example 42 includes the subject matter of any of Examples 37-41, and wherein the means for adjusting the batch size comprises means for decreasing the batch size for a processing stage of the network device.
[0100] Example 43 includes the subject matter of any of Examples 37-42, and wherein the means for adjusting the batch size comprises means for additively decreasing the batch size for a processing stage of the network device.
[0101] Example 44 includes the subject matter of any of Examples 37-43, and further including circuitry for processing one or more packets with multiple processing stages, wherein each processing stage performs a different operation on packet data.
[0102] Example 45 includes the subject matter of any of Examples 37-44, and further including means for synchronizing the adjusted batch size across multiple processing stages of the network device.

[0103] Example 46 includes the subject matter of any of Examples 37-45, and wherein the means for synchronizing the adjusted batch size comprises means for comparing the batch size of a present processing stage with the batch size for every preceding processing stage in the network device; and means for adjusting, in response to a determination that the batch size of the present processing stage is greater than the batch size for every preceding processing stage, the batch size for every preceding processing stage to the batch size of the present processing stage.
[0104] Example 47 includes the subject matter of any of Examples 37-46, and further including means for comparing the batch size of a present processing stage with a batch size for a preceding processing stage; and means for maintaining, in response to a determination that the batch size of the present processing stage is smaller than the batch size for the preceding processing stage, the batch size of the preceding processing stage.

Claims

1. Network switch circuitry to be comprised in a network switch, the network switch being configurable to be communicatively coupled to at least one other network switch in at least one network, the network switch circuitry comprising: network communication circuitry for use in data communications with the at least one other network switch via the at least one network; and circuitry to perform operations comprising: queuing received packet data for use in packet data transmissions associated with one or more transmission sources and one or more transmission destinations in the at least one network, the one or more transmission sources and/or the one or more transmission destinations being associated, at least in part, with the network switch and/or the at least one other network switch; dynamically determining a packet data transmission batch size to be applied in one of the packet data transmissions; after the packet data batch size has been determined, generating a batch of queued packet data in accordance with the packet data transmission batch size; and dynamically determining whether to adjust the batch size to provide an adjusted batch size to be applied to additional queued data in at least one other of the packet data transmissions; wherein: the at least one network comprises one or more respective graphics processing units (GPUs) that are associated with the one or more transmission sources and/or the one or more transmission destinations; the dynamically determining the packet data transmission batch size and the dynamically determining whether to adjust the batch size are to be carried out so as to permit batch size coordination across multiple of the packet data transmissions; the dynamically determining the packet data transmission batch size, the dynamically determining whether to adjust the batch size, and/or the packet data transmissions are to be based, at least in part, upon: congestion-related information associated with the packet data transmissions; and/or queuing condition information associated with the received data.
2. The network switch circuitry of claim 1, wherein: an integrated circuit chip comprises the network switch circuitry.
3. The network switch circuitry of claim 2, wherein: a system-on-chip comprises the network switch circuitry.
4. The network switch circuitry of claim 3, wherein: the adjusted batch size is increased or decreased relative to the batch size.
5. The network switch circuitry of claim 4, wherein: the batch size coordination is implemented across multiple processing nodes associated with the multiple of the packet data transmissions.
6. The network switch circuitry of claim 5, wherein: the multiple processing nodes are associated, at least in part, with multiple processing stages; and the batch size coordination comprises batch size synchronization across the multiple processing stages.
7. The network switch circuitry of claim 4, wherein: the batch size coordination is to avoid packet data loss and/or packet data dropping.
8. A method implemented using network switch circuitry that is to be comprised in a network

switch, the network switch being configurable to be communicatively coupled to at least one other network switch in at least one network, the network switch circuitry comprising network communication circuitry, the method comprising: using the network communication circuitry in data communications with the at least one other network switch via the at least one network; queuing, by the network switch circuitry, received packet data for use in packet data transmissions associated with one or more transmission sources and one or more transmission destinations in the at least one network, the one or more transmission sources and/or the one or more transmission destinations being associated, at least in part, with the network switch and/or the at least one other network switch; dynamically determining, by the network switch circuitry, a packet data transmission batch size to be applied in one of the packet data transmissions; after the packet data batch size has been determined, generating, by the network switch circuitry, a batch of queued packet data in accordance with the packet data transmission batch size; and dynamically determining, by the network switch circuitry, whether to adjust the batch size to provide an adjusted batch size to be applied to additional queued data in at least one other of the packet data transmissions; wherein: the at least one network comprises one or more respective graphics processing units (GPUs) that are associated with the one or more transmission sources and/or the one or more transmission destinations; the dynamically determining the packet data transmission batch size and the dynamically determining whether to adjust the batch size are to be carried out so as to permit batch size coordination across multiple of the packet data transmissions; the dynamically determining the packet data transmission batch size, the dynamically determining whether to adjust the batch size, and/or the packet data transmissions are to be based, at least in part, upon: congestion-related information associated with the packet data transmissions; and/or queuing condition information associated with the received data.
9. The method of claim 8, wherein: an integrated circuit chip comprises the network switch circuitry.
10. The method of claim 9, wherein: a system-on-chip comprises the network switch circuitry.
11. The method of claim 10, wherein: the adjusted batch size is increased or decreased relative to the batch size.
12. The method of claim 11, wherein: the batch size coordination is implemented across multiple processing nodes associated with the multiple of the packet data transmissions.
13. The method of claim 12, wherein: the multiple processing nodes are associated, at least in part, with multiple processing stages; and the batch size coordination comprises batch size synchronization across the multiple processing stages.
14. The method of claim 11, wherein: the batch size coordination is to avoid packet data loss and/or packet data dropping.
15. At least one non-transitory machine-readable storage medium storing instructions for being executed by network switch circuitry that is to be comprised in a network switch, the network switch being configurable to be communicatively coupled to at least one other network switch in at least one network, the network switch circuitry comprising network communication circuitry, the instructions, when executed by the network switch circuitry, resulting in the network switch circuitry being configured for performance of operations comprising: using the network communication circuitry in data communications with the at least one other network switch via the at least one network; queuing, by the network switch circuitry, received packet data for use in packet data transmissions associated with one or more transmission sources and one or more transmission destinations in the at least one network, the one or more transmission sources and/or the one or more transmission destinations being associated, at least in part, with the network switch and/or the at least one other network switch; dynamically determining, by the network switch circuitry, a packet data transmission batch size to be applied in one of the packet data transmissions; after the packet data batch size has been determined, generating, by the network switch circuitry, a batch of queued packet data in accordance with the packet data transmission

batch size; and dynamically determining, by the network switch circuitry, whether to adjust the batch size to provide an adjusted batch size to be applied to additional queued data in at least one other of the packet data transmissions; wherein: the at least one network comprises one or more respective graphics processing units (GPUs) that are associated with the one or more transmission sources and/or the one or more transmission destinations; the dynamically determining the packet data transmission batch size and the dynamically determining whether to adjust the batch size are to be carried out so as to permit batch size coordination across multiple of the packet data transmissions; the dynamically determining the packet data transmission batch size, the dynamically determining whether to adjust the batch size, and/or the packet data transmissions are to be based, at least in part, upon: congestion-related information associated with the packet data transmissions; and/or queuing condition information associated with the received data.

16. The at least one non-transitory machine-readable storage medium of claim 15, wherein: an integrated circuit chip comprises the network switch circuitry.

17. The at least one non-transitory machine-readable storage medium of claim 16, wherein: a system-on-chip comprises the network switch circuitry.

18. The at least one non-transitory machine-readable storage medium of claim 17, wherein: the adjusted batch size is increased or decreased relative to the batch size.

19. The at least one non-transitory machine-readable storage medium of claim 18, wherein: the batch size coordination is implemented across multiple processing nodes associated with the multiple of the packet data transmissions.

20. The at least one non-transitory machine-readable storage medium of claim 19, wherein: the multiple processing nodes are associated, at least in part, with multiple processing stages; and the batch size coordination comprises batch size synchronization across the multiple processing stages.

21. The at least one non-transitory machine-readable storage medium of claim 18, wherein: the batch size coordination is to avoid packet data loss and/or packet data dropping.

22. A network system comprising: multiple graphics processing units (GPUs) associated with multiple transmission sources and/or multiple transmission destinations in at least one network; multiple network switches communicatively coupled together via the at least one network, the multiple transmission sources and/or the multiple transmission destinations being associated, at least in part, with the multiple network switches, the multiple network switches comprising at least one network switch and at least one other network switch, the at least one network switch comprising network switch circuitry, the network switch circuitry to perform operations comprising: queuing received packet data for use in packet data transmissions associated, at least in part, with the multiple transmission sources and the multiple transmission destinations in the at least one network; dynamically determining a packet data transmission batch size to be applied in one of the packet data transmissions; after the packet data batch size has been determined, generating a batch of queued packet data in accordance with the packet data transmission batch size; and dynamically determining whether to adjust the batch size to provide an adjusted batch size to be applied to additional queued data in at least one other of the packet data transmissions; wherein: the dynamically determining the packet data transmission batch size and the dynamically determining whether to adjust the batch size are to be carried out so as to permit batch size coordination across multiple of the packet data transmissions; the dynamically determining the packet data transmission batch size, the dynamically determining whether to adjust the batch size, and/or the packet data transmissions are to be based, at least in part, upon: congestion-related information associated with the packet data transmissions; and/or queuing condition information associated with the received data.

23. The network system of claim 22, wherein: an integrated circuit chip comprises the network switch circuitry.

24. The network system of claim 23, wherein: a system-on-chip comprises the network switch circuitry.

25. The network system of claim 24, wherein: the adjusted batch size is increased or decreased relative to the batch size.

26. The network system of claim 25, wherein: the batch size coordination is implemented across multiple processing nodes associated with the multiple of the packet data transmissions.

27. The network system of claim 26, wherein: the multiple processing nodes are associated, at least in part, with multiple processing stages; and the batch size coordination comprises batch size synchronization across the multiple processing stages.

28. The network system of claim 25, wherein: the batch size coordination is to avoid packet data loss and/or packet data dropping.
